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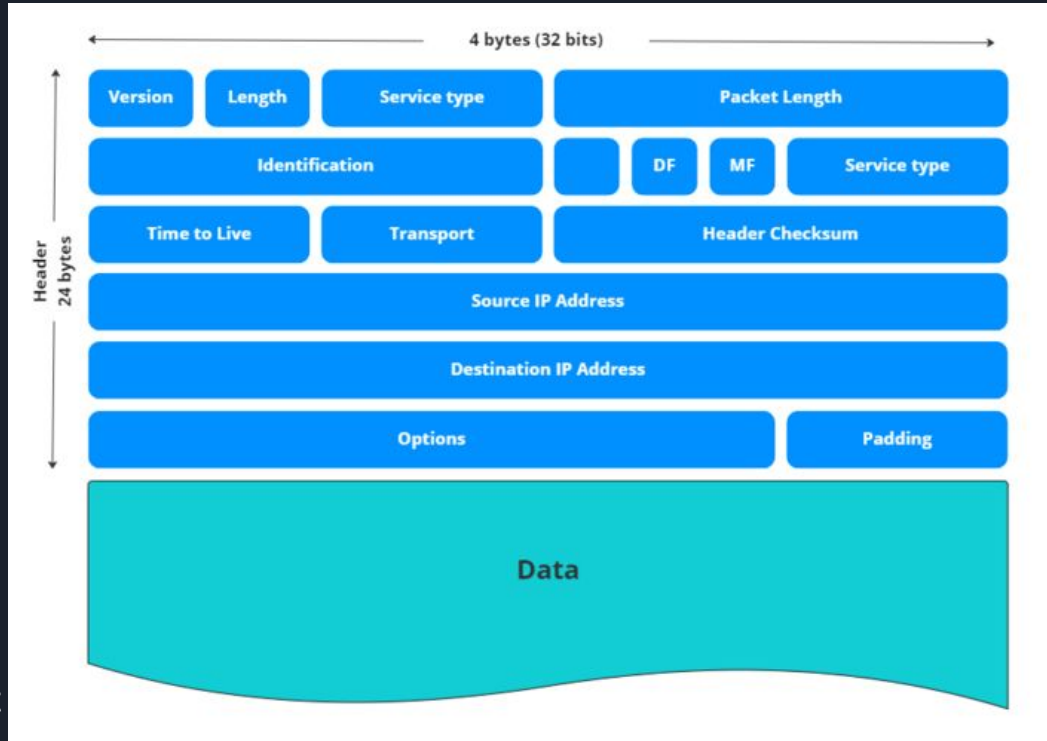
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What is an IP Packets ?

A packet is the smallest unit of data sent over a network.

In most computer networks, data is broken down into small units called packets and sent between devices. The same holds true on the internet. Each packet has a specific structure, defined by a protocol, that allows the recipient to understand the packet's contents. An IP packet is a unit of data in a network that contains information about the source and destination addresses and other control information needed to transport the packet over a network.





IP/Data Packets

Data is broken into **packets**, transmitted, then reassembled.

Every packet has:

[Headers] + [Payload]

Headers = control info

Payload = actual data

Each packet contains both **data** and **control information**. The data portion (payload) contains the actual content being transmitted, such as text, images, or video.

While the control information in the packet header includes details such as the source and destination IP addresses, time to live, packet length, and routing instructions.

Packet Encapsulation :

Data moves through layers:

Application → Transport → Network → Data Link

Each layer adds its own header:

[Ethernet] [IP] [TCP] [Data]

This is called **encapsulation**.



Structure of an IP packet

Several fields in an IP packet header help route and deliver the packet to its desired destination.

- **Version:** This field indicates the IP protocol version, such as IPv4 or IPv6, which helps determine how the packet should be routed.
- **Total length:** Specifies the packet's total length of the data payload (the message) and the header, which are essential for the recipient to understand the packet's contents.
- **Protocol:** Determines the high-level protocol, like TCP, UDP, etc., used in the data payload.
- **Time to Live (TTL):** Used to prevent packets from circulating indefinitely, which could cause network congestion. Every time a packet gets forwarded, the router decreases the TTL by one, and when the TTL becomes zero, the packet gets discarded, helping to avoid congestion.
- **Source IP address:** Specifies the IP address of the device that sent the packet. It's used to reply to the sender, and if a router cannot send the packet to the following location, it sends a message to the source using the source IP address to indicate the problem.
- **Destination IP address:** Identifies the IP address of the intended recipient of the packet. Routers use it to determine where to forward the packet to reach its final destination.



IPv4 Header Fields - Layer 3 Network Layer

Field	Meaning	SOC Use
Version	IPv4 or IPv6	Protocol identification
Source IP	Sender	Identify attacker
Destination IP	Receiver	Target system
TTL (Time To Live)	Packet lifetime	Detect anomalies
Protocol	TCP/UDP/ICMP	Traffic type
Total Length	Packet size	Detect exfiltration
Header Checksum	Integrity	Corruption detection



TCP Header : Layer 4

Field	Meaning	SOC Use
Source Port	Sender port	Identify origin
Destination Port	Target service	Detect attack
Sequence Number	Order tracking	Session analysis
ACK Number	Confirmation	Flow validation
Flags	Connection state	Attack detection
Window Size	Flow control	Traffic behavior



Example Packet

Source IP: 192.168.1.10
Destination IP: 45.77.123.66
Protocol: TCP
TTL: 64
Length: 1500 bytes

SOC Analysis

Source IP

- Internal → normal
- External → suspicious

Destination IP

- Known service → OK
- Unknown / malicious → alert

TTL - Time to live

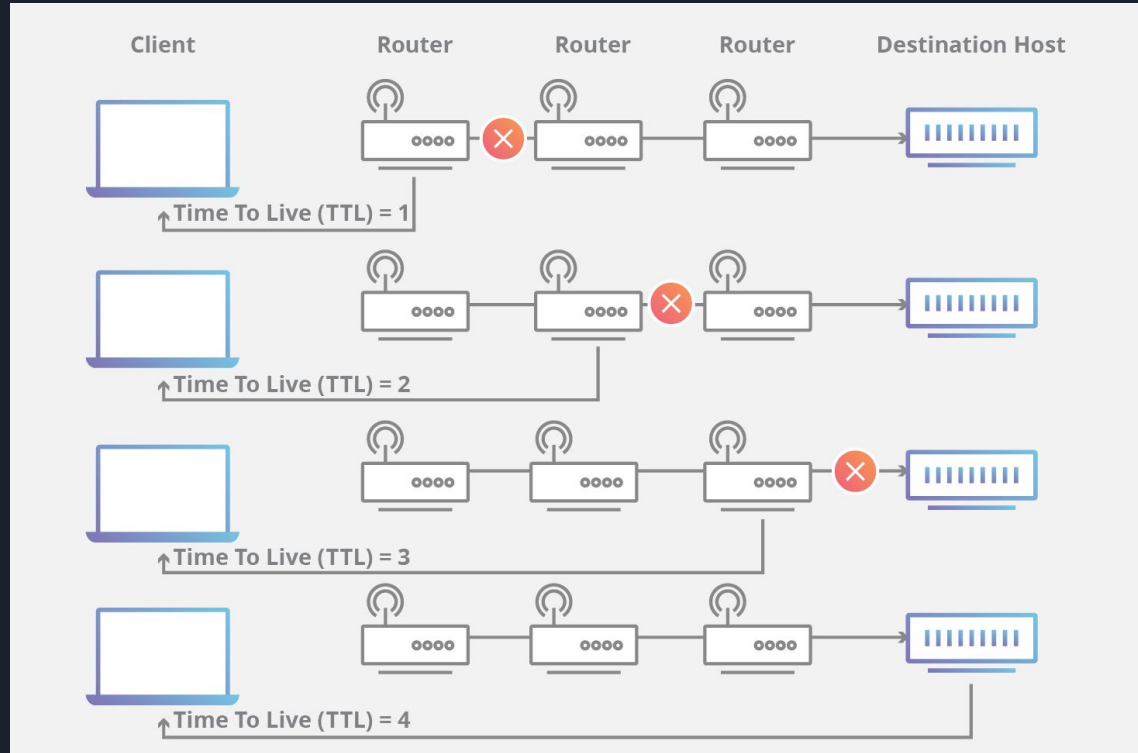


Image Credit : cloudflare.com

What is TTL - Time To Live

Time to live (TTL) refers to the amount of time or “hops” that a packet is set to exist inside a network before being discarded by a router.

TTL is also used in other contexts including CDN caching and DNS caching.





How TTL Works

When a packet of information is created and sent out across the Internet, there is a risk that it will continue to pass from router to router indefinitely.

To mitigate this possibility, packets are designed with an expiration called a time-to-live or hop limit.

Packet TTL can also be useful in determining how long a packet has been in circulation, and allow the sender to receive information about a packet's path through the Internet.

Each packet has a place where it stores a numerical value determining how much longer it should continue to move through the network.

Every time a router receives a packet, it subtracts one from the TTL count and then passes it onto the next location in the network.

If at any point the TTL count is equal to zero after the subtraction, the router will discard the packet and send an ICMP message back to the originating host.



TTL - Time to live

How TTL Works

The commonly used network commands ping and traceroute both utilize TTL. When using the traceroute command, a stream of packets with increasingly higher sequential TTLs are sent across the Internet towards a destination.

Because each step along the connection is the last stop for one of the packets, each location will return an ICMP message to the sender after discarding the packet.

The time it takes for the ICMP message to return to the sender is then used to determine how long it takes to get to each successive hop along the network.

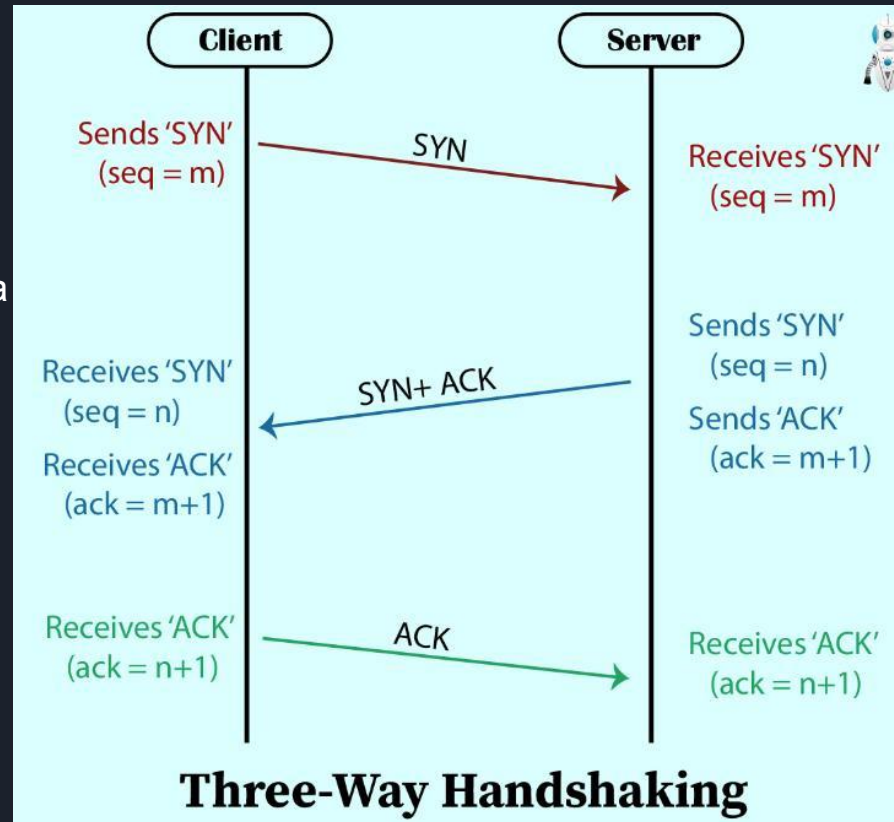
3 Way Handshake



TCP 3 Way Handshake

The handshake is a data-sharing process that involves a back-and-forth communication between a client (for example a mobile phone or Smart TV) and a server.

The purpose of this communication is to establish a secure and reliable connection so that data (files, images, videos, etc.) can be transferred efficiently between the systems.





3 Way Handshake

TCP Flags :

Flag	Meaning	SOC Use
SYN	Start	Connection attempts
ACK	Confirm	Normal traffic
FIN	Close	Session end
RST	Reset	Abnormal termination
PSH	Push data	Immediate delivery
URG	Urgent	Rare

Client → SYN
Server → SYN-ACK
Client → ACK



Connection established

3 Way Handshake



Suspicious Patterns For SOC

Pattern	Meaning
SYN only	SYN flood
SYN + FIN	Malicious / scan
RST flood	Disruption
No ACK	Half-open attack



3 Way Handshake

Real SOC Scenario

Source IP: 185.221.x.x

Packets: 100,000 SYN

ACK: 0

Destination: Port 80

Conclusion:

SYN Flood DDoS